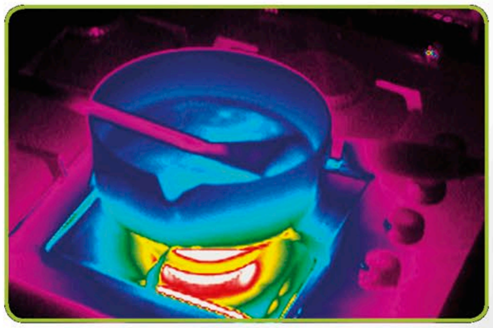
Metals are good heat conductors because their electrons are free to move and pass energy on.

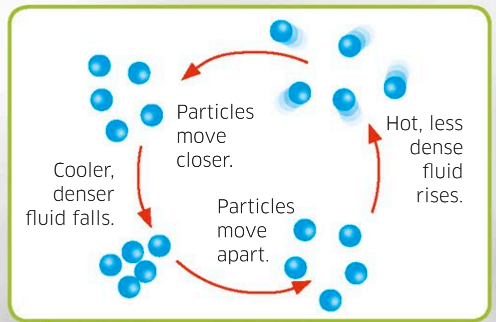
Copper, gold, silver, and aluminum are all good conductors of heat.

Heat transfer

Heat in this pan of boiling water can be seen to move in three ways—radiation, conduction, and convection—between the heat source, the metal pan, and the water.



Heat distribution
A thermogram (infrared image) reveals how heat is distributed from the hottest point, the flame, to the coldest, the wooden spoon and stove.



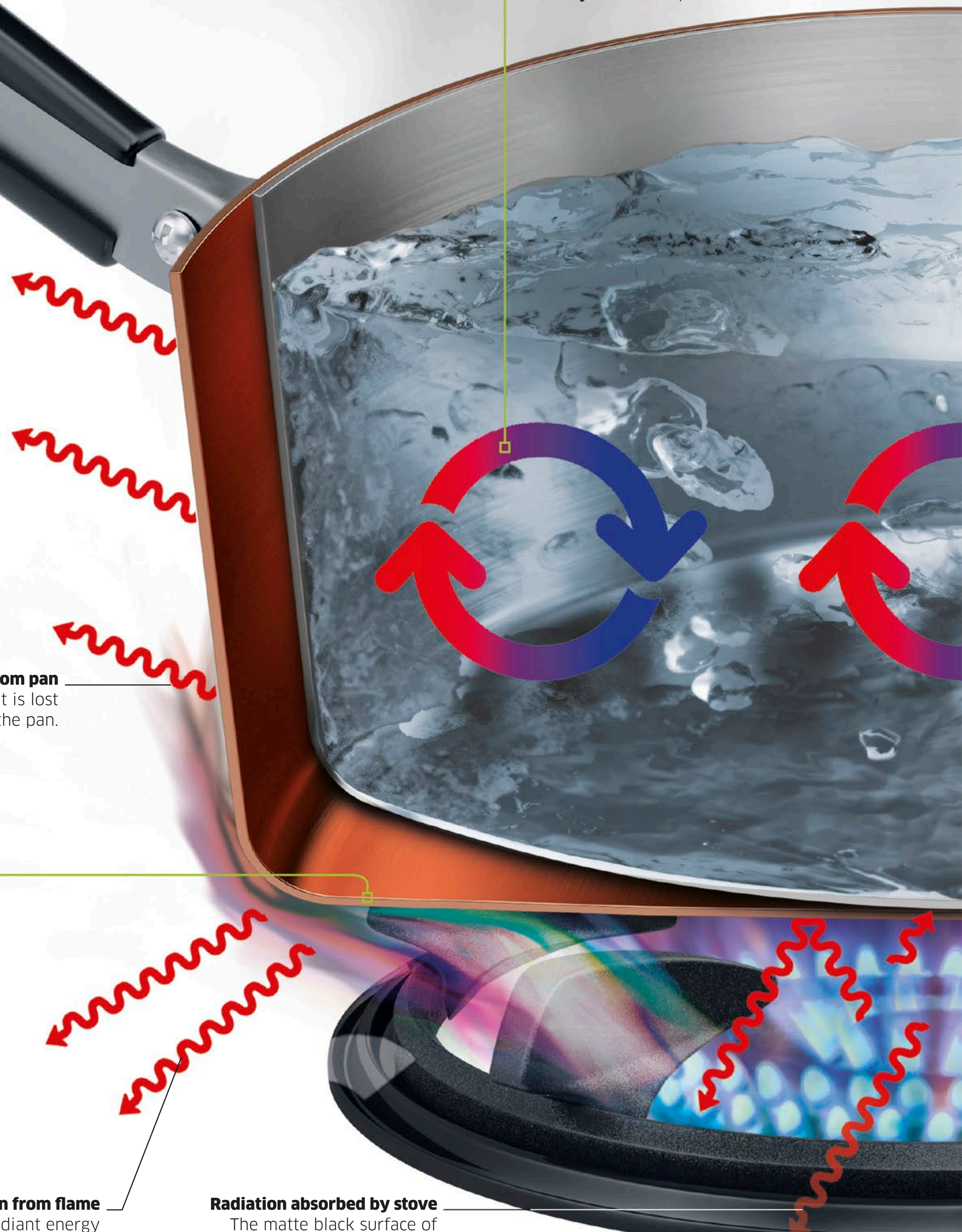
Convection
As a fluid (liquid or gas) heats up, the particles of which it is made move apart, so the fluid becomes less dense and rises. As it moves away from the heat source, the fluid cools down, its density increases, and it falls.

Thermal insulation
Materials such as plastic and wood are thermal insulators, which do not conduct heat.

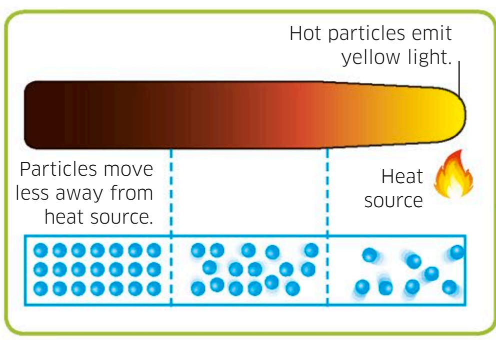
Heat

Heat is energy that increases the temperature of a substance or makes it change state—from a liquid to a gas, for example. Heat can move into or within a substance in three ways: conduction, convection, or radiation.

Atoms and molecules are always moving around. The energy of their movement is called kinetic energy. Some move faster than others, and the temperature of a substance is the average kinetic energy of its atoms and molecules.



Radiation from pan
Some radiant heat is lost from the side of the pan.



Conduction
When the particles (atoms or molecules) of a solid are heated, they move faster, bumping into other particles and making them move faster, too. The movement of the particles conducts heat away from the heat source. As the temperature increases in a metal, the particles lose heat as thermal radiation, making the metal glow red, yellow, and then white hot.

Radiation from flame
Heat moves as radiant energy waves through a gas or vacuum. This is how the sun heats Earth.

Radiation absorbed by stove
The matte black surface of the stove absorbs some heat radiation.